ACE Semantic Gateway: Geometric Stabilization of Language Model Inference.

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Executive Summary

Problem

Large Language Models (LLMs) generate responses by exploring high-dimensional semantic spaces learned from training data.
While this probabilistic approach enables impressive language generation, it also introduces a persistent problem: **semantic instability**.

When prompts contain ambiguity, incomplete framing, or hidden assumptions, models may explore multiple competing interpretations. This often produces:

- hallucinated responses
- unstable reasoning chains
- excessive token generation
- increased computational cost

Current mitigation techniques typically operate **after generation**, using filtering, retrieval, or secondary model evaluation.

Core Idea

The ACE (Axiomatic Contextual Evaluation) framework proposes a different approach:

stabilize the semantic context before generation occurs.

ACE introduces a semantic middleware layer that analyzes incoming requests, constructs a contextual reference field, and evaluates candidate responses relative to that field before invoking the language model.

This shifts the problem from **probabilistic correction** to **geometric semantic alignment**.

Key Insight

Human language varies across expressions, but the **underlying meaning remains structurally invariant**.

Modern language models represent meaning as vectors in high-dimensional embedding spaces. Within these spaces:

- semantic relations form stable geometric structures
- contextual domains correspond to low-dimensional subspaces
- coherent interpretations lie near these structures

ACE leverages this geometric structure of meaning.

The ACE Criterion

ACE evaluates semantic alignment using the **Origin Cost**:

$$
O(z) = ||z - P_S(z)||^2
$$

Where:

- (z) = vector representation of a candidate response
- (S) = contextual subspace defined by domain invariants
- (P_S(z)) = projection onto that subspace

The Origin Cost measures the deviation between a response and the contextual semantic structure.

Interpretation:

low origin cost → semantic alignment
high origin cost → semantic drift

Context Matrix

The contextual subspace is constructed using the **ACE Context Matrix**, which encodes the structural elements of the domain:

Domain Anchors
Semantic Invariants
Constraint Rules

In the middleware prototype, these invariants were initially expressed using explicit semantic terms for interpretability.

However, once embedded in vector space, ACE operates directly on their **vector representations**, allowing contextual geometry to be defined without reliance on symbolic ontologies.

Semantic Energy Interpretation

Within the ACE framework, the Origin Cost can be interpreted as **semantic potential energy**.

Coherent interpretations correspond to **low-energy states** within the contextual field.

These low-energy regions form **semantic attractors**, toward which stable interpretations converge.

By restricting inference to these attractors, ACE stabilizes semantic reasoning.

Low-Dimensional Context Hypothesis

Although embedding spaces may contain thousands of dimensions, coherent contextual reasoning typically occurs within **low-dimensional subspaces** defined by domain invariants.

This drastically reduces the effective search space explored during inference.

Consequences

Constraining inference to contextual semantic subspaces produces several benefits:

- reduced semantic ambiguity
- fewer competing interpretations
- shorter reasoning paths
- lower token usage
- reduced computational energy consumption

Architectural Implementation

The ACE Semantic Gateway introduces a stabilization layer before language generation:

Input
↓
Context Matrix Construction
↓
Contextual Subspace Definition
↓
Origin Cost Evaluation
↓
Decision Layer (Answer / Clarify / Abstain)
↓
LLM Generation

This architecture separates **semantic interpretation** from **language generation**, improving reliability without modifying the underlying model.

Conceptual Contribution

ACE reframes hallucination not simply as a training artifact, but as a form of **geometric instability in semantic representation space**.

By aligning responses with invariant semantic structures, ACE introduces a mechanism for stabilizing inference in language models.

Broader Perspective

Because embeddings represent meaning independently of specific linguistic expressions, ACE operates at the level of **semantic structure rather than language form**.

This suggests a path toward **universal semantic alignment**, where communication can be stabilized across different languages and linguistic expressions by operating directly on the geometry of meaning.

One-sentence summary

> ACE stabilizes language model inference by aligning candidate responses with invariant semantic structures in vector space, transforming semantic coherence from a probabilistic property into a geometric criterion.

ACE Context Matrix Specification (Draft v1)

1. Objective

The ACE Context Matrix defines the **semantic structure of an active context** before language generation.

Its purpose is to construct a **stable semantic reference field** that allows the ACE Origin Cost criterion to evaluate candidate responses.

The Context Matrix determines:

- which semantic domain is active

- which invariants govern that domain
- which vector subspace defines the contextual geometry

---

2. Conceptual Architecture

The ACE Context Matrix is composed of three structural layers:

```
Context Matrix
├── Domain Anchors
├── Semantic Invariants
└── Constraint Rules
```

Each layer contributes to the construction of a contextual reference subspace.

---

3. Domain Anchors

Domain anchors are **semantic reference vectors** that identify the conceptual region of the input.

They correspond to stable semantic concepts extracted from the input query.

Example:

fuel
distance
consumption
vehicle
tank

Each anchor is represented as an embedding vector:

$$
v_i \in \mathbb{R}^n
$$

These anchors define the **initial semantic region** of the problem.

---

4. Semantic Invariants

Semantic invariants represent **structural relations that remain stable within the domain**.

Examples include:

fuel enables motion
consumption relates fuel and distance
distance ≥ 0
fuel ≥ 0

In vector representation, invariants correspond to **directional relations** between anchors.

Example:

[
d_1 = v_{distance} - v_{fuel}
]

These invariant directions define the **structural geometry of the domain**.

---

5. Constraint Rules

Constraint rules define **logical or physical restrictions** that limit semantic interpretation.

Examples:

distance cannot be negative
fuel consumption must be positive
cause precedes effect

Constraints serve two roles:

1. restrict the valid contextual subspace
2. detect semantic contradictions

---

6. Construction of the Contextual Subspace

Once anchors and invariants are defined, the contextual subspace is constructed.

Let:

$$
S = \mathrm{span}(v_1, v_2, \ldots, v_k, d_1, d_2, \ldots, d_m)
$$

Where:

- v_i are anchor vectors
- d_j are invariant directions

The resulting subspace S represents the **semantic geometry of the active context**.

In practice, the effective dimension of S is typically much smaller than the full embedding space.

Example:

embedding space: 1536 dimensions
context subspace: 10–50 dimensions

---

7. Contextual Reference Field

The contextual reference field is the geometric region defined by the contextual subspace S.

All semantically coherent responses should lie approximately within this field.

Candidate responses are represented as vectors:

$$
z \in \mathbb{R}^n
$$

---

8. Origin Cost Evaluation

ACE evaluates the semantic stability of a candidate response using the Origin Cost:

$$
O(z) = \|z - P_S(z)\|^2
$$

Where:

- $(P_S(z))$ is the orthogonal projection of (z) onto the contextual subspace (S)

Interpretation:

$O(z) \approx 0$ → response aligned with context
$O(z)$ large → semantic drift

---

9. Decision Layer

Based on the Origin Cost, the ACE system determines the appropriate action:

Low Origin Cost → Answer
Medium Origin Cost → Clarify
High Origin Cost → Abstain

This decision occurs **before invoking the language model**.

---

10. Semantic Energy Interpretation

The Origin Cost can be interpreted as **semantic potential energy**.

Stable interpretations correspond to **low-energy states**.

Hallucinations correspond to **high-energy deviations from the contextual field**.

---

11. Implications

The Context Matrix enables a new architecture for LLM systems:

Input
↓
Context Matrix Construction
↓
Contextual Subspace Definition
↓
Origin Cost Evaluation

↓
Decision
↓
LLM Generation

This architecture reduces semantic entropy by restricting inference to a **stable contextual geometry**.

---

12. Expected Effects

Constraining inference within contextual subspaces can produce:

- reduced semantic exploration
- lower token usage
- improved coherence
- lower computational energy consumption

---

13. Future Work

Future research directions include:

- automated extraction of invariants
- dynamic context matrix construction
- integration with retrieval systems
- empirical measurement of semantic energy landscapes

Semantic Attractors and Energy Landscapes

Interpreting ACE as an energy-based semantic stabilization mechanism

1. Semantic inference as a search process

Language model generation can be interpreted as a **search through a high-dimensional semantic space**.

At each step of generation, the model evaluates multiple candidate continuations based on probability distributions derived from training data.

This process can be conceptualized as exploration within a **semantic landscape**, where each possible continuation corresponds to a point in representation space.

In unstable contexts, the model may explore many competing semantic regions, producing:

- long reasoning chains
- token proliferation
- hallucinated responses

This behavior corresponds to **high semantic entropy**.

---

2. Energy interpretation of semantic deviation

Within the ACE framework, semantic stability is evaluated using the Origin Cost function:

O(z)= // z−PS(z) // 2O(z) = \|z - P_S(z)\|^2O(z)= // z−PS(z) // 2

This function measures the distance between a candidate response zzz and the contextual reference subspace SSS.

This distance can be interpreted as **semantic potential energy**.

In this interpretation:

low origin cost → low semantic energy
high origin cost → high semantic energy

Semantically coherent responses occupy **low-energy regions of the contextual field**.

---

3. Formation of semantic attractors

When a contextual subspace is defined by domain invariants, the resulting geometry introduces **stable regions in semantic space**.

These regions act as **semantic attractors**.

An attractor is a region toward which a dynamic system tends to converge.

In the ACE framework:

semantic invariants → define subspace geometry
subspace geometry → defines energy minima
energy minima → act as semantic attractors

These attractors represent **stable interpretations of the input problem**.

---

4. Suppression of unstable interpretations

Without contextual stabilization, the semantic landscape may contain multiple shallow minima corresponding to ambiguous interpretations.

This produces unstable generation paths.

The ACE architecture suppresses these unstable regions by evaluating semantic deviation before generation.

High-energy interpretations are rejected before they propagate into language generation.

As a result, inference remains near the contextual attractors.

---

5. Reduction of semantic exploration

When inference is constrained near semantic attractors, the model no longer needs to explore large regions of the embedding space.

The effective search region becomes limited to a **low-dimensional contextual manifold**.

This produces three effects:

smaller semantic search space
↓
fewer competing interpretations
↓
shorter generation paths

---

6. Relationship to token usage

Token generation in language models is closely related to the depth of semantic exploration.

When multiple interpretations compete, the model generates longer reasoning sequences to resolve ambiguity.

By stabilizing the semantic context before generation, ACE reduces ambiguity and narrows the search region.

This reduces:

- token usage
- inference time
- computational energy consumption

---

7. Relation to energy-based models

The ACE framework can be interpreted as a form of **energy-based semantic inference**.

Energy-based models assign energy values to candidate states and select low-energy solutions.

In ACE:

semantic candidate → vector z
energy → origin cost O(z)
solution → minimal energy state

This perspective aligns ACE with established principles in energy-based machine learning.

---

8. Contextual stabilization before generation

Most existing techniques attempt to correct hallucinations **after generation**.

Examples include:

- output filtering
- secondary model evaluation
- retrieval augmentation

The ACE architecture introduces stabilization **before generation**.

The Context Matrix defines the semantic field, and the Origin Cost evaluates alignment within that field.

Only responses compatible with the contextual geometry are allowed to propagate into generation.

---

9. Implications for AI architecture

The attractor interpretation suggests a new design paradigm for language model systems.

Instead of treating language models as autonomous reasoning engines, they can be integrated within a **stabilized semantic environment**.

This architecture separates two functions:

semantic stabilization
language generation

The first constrains the semantic space, while the second produces linguistic output.

---

10. Consequences for computational efficiency

Because ACE reduces semantic exploration, the model performs less probabilistic branching during generation.

This has direct implications for efficiency:

- fewer tokens generated
- reduced inference computation
- lower energy consumption in large-scale deployments

This observation aligns with empirical evidence that stabilizing semantic context can significantly reduce token usage.

---

11. Summary

The ACE architecture introduces a geometric stabilization layer before language generation.

By defining contextual subspaces and evaluating semantic deviation using an energy criterion, the system creates **semantic attractors** that guide inference toward stable interpretations.

This approach reframes hallucination as a form of **geometric instability in semantic representation space**, and proposes stabilization as a method for improving both reliability and efficiency in language model systems.

Diagrama conceptual del ACE Semantic Gateway

La arquitectura se puede visualizar como **tres capas principales**.

```
INPUT LAYER
User Query
   │
   ▼
SEMANTIC INTERPRETATION
(Context Matrix Construction)
   │
   ├── Domain Anchors
   ├── Semantic Invariants
   └── Constraint Rules
   │
   ▼
CONTEXTUAL SUBSPACE
(Semantic Reference Field S)
   │
   ▼
ACE CRITERION
Origin Cost Evaluation
O(z) = || z - P_S(z) ||²
   │
   ▼
DECISION LAYER
   ├── Low Cost → Answer
   ├── Medium Cost → Clarify
   └── High Cost → Abstain
   │
   ▼
LLM GENERATION
Constrained Language Output
```

Diagrama geométrico (la parte más importante)

Este es el que explica **la idea matemática**.

High-Dimensional Embedding Space (R^n)

```
                z (candidate response)
                  *
                 /
                /
               /
              /
--------------/------------------------
             /|
            / |
           /  |
          /   |
         /    |
        /     |
       /      |
      /       |
     /        |
    /         |
   /          |
  S (Contextual Subspace)
```

Interpretación:

Projection onto contextual subspace

P_S(z)

Origin Cost

O(z) = distance²(z , S)

Diagrama de energía semántica

Este conecta con la reducción de tokens.

Semantic Energy Landscape

```
High Energy
   ▲
   |
   |
   |      unstable interpretations
   |     /\   /\  /\
   |    /  \ /  \ /  \
   |
   |
   |
   |
   |
   |_______/\_________/\_________
         attractor

  stable semantic interpretation

      Low Energy
```

ACE restringe la búsqueda a los **valles estables**.

Diagrama del flujo completo (el que recomiendo usar)

```
┌──────────────────────────┐
│     User Query           │
└──────────┬───────────────┘
           │
           ▼
┌──────────────────────────────┐
│  Context Matrix              │
│                              │
│ Domain Anchors               │
│ Semantic Invariants          │
│ Constraint Rules             │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
```

```
│ Contextual Subspace S          │
│ (Semantic Reference)           │
└────────────┬────────────────────┘
             │
             ▼
┌────────────────────────────────┐
│ ACE Origin Cost                │
│ O(z) = ||z - P_S(z)||²         │
└────────────┬────────────────────┘
             │
             ▼
┌────────────────────────────────┐
│ Decision Layer                 │
│                                │
│ Answer                         │
│ Clarify                        │
│ Abstain                        │
└────────────┬────────────────────┘
             │
             ▼
┌────────────────────────────────┐
│ Language Model                 │
│ Generation                     │
└────────────────────────────────┘
```

Figura 1 — ACE Semantic Stabilization Framework

Conceptualmente la figura debe tener **tres zonas**:

1 Espacio semántico completo
2 Subespacio contextual
3 Evaluación del criterio ACE

La estructura sería así:

Figure 1
ACE Semantic Stabilization Framework

High-Dimensional Embedding Space ($\mathbb{R}^n$)

```
        candidate response z
                ●
               /
              /
             /
            /
           /
          /
         /
        /
       /
      /
     /
    /      Contextual Subspace S
   /       ____________________________
  /      /                    /
 /      /                    /
/      /                    /
/      /                    /
/      /                    /
/      /                    /
/________/____________________________/_________
```

Projection onto contextual field

P_S(z)

Origin Cost

O(z) = || z − P_S(z) ||²

Interpretación:

Low Origin Cost → semantic alignment
High Origin Cost → semantic drift

Cómo debe verse gráficamente

La figura ideal tendría:

lado izquierdo
→ espacio vectorial grande

centro
→ plano o subespacio contextual

punto fuera del plano
→ respuesta candidata

flecha
→ proyección

distancia
→ costo de origen

Caption de la figura (esto es importante para el paper)

Puedes usar algo así:

Figure 1. Semantic stabilization through contextual subspaces.
Large language model responses are represented as vectors in a high-dimensional embedding space. The ACE Context Matrix defines a contextual reference subspace based on domain anchors and semantic invariants. Candidate responses are evaluated using the Origin Cost (O(z)), defined as the squared distance between the response vector and its projection onto the contextual subspace. Low Origin Cost corresponds to semantically coherent responses, while large deviations indicate semantic instability.

Ahora vamos a la explicación que quieres agregar

Esta es la parte clave que mencionaste:

> explicar que ACE es la construcción de una matriz de criterio basada en la estructura semántica.

Voy a escribirlo en forma clara y científica.

ACE as a Semantic Criterion Matrix

The ACE framework can be interpreted as the construction of a **semantic criterion matrix** derived from the structural properties of language.

Human language encodes meaning through stable relational structures such as causality, physical constraints, and domain-specific dependencies. These structures generate **semantic invariants** that can be represented geometrically in embedding spaces.

Within this representation, semantic concepts correspond to vectors in a high-dimensional space, while invariant relations define **stable directions and subspaces**.

The ACE Context Matrix formalizes this structure by combining:

domain anchors
semantic invariants
constraint rules

These elements define a contextual reference subspace that represents the **semantic geometry of the active domain**.

Candidate responses generated by a language model can then be evaluated relative to this structure.

The ACE Origin Cost measures the deviation between a candidate response vector and the contextual reference field:

$$
O(z) = \| z - P_S(z) \|^2
$$

This formulation transforms semantic coherence into a **geometric criterion**.

Instead of relying solely on probabilistic token prediction, the system evaluates whether a response remains aligned with the contextual semantic structure.

Consequence: token optimization

When inference is constrained within a contextual subspace defined by semantic invariants, the model explores a significantly smaller region of the representation space.

This produces two important effects:

1. reduced semantic entropy during inference
2. reduced token exploration during generation

As a result, the system requires fewer tokens to reach a coherent response and avoids unstable reasoning paths.

In this sense, ACE can be interpreted as a **semantic optimization layer** that improves both response reliability and computational efficiency.

Figure 1
ACE semantic stabilization

Section
Context Matrix Specification

Section
Origin Cost Criterion

Section
Semantic Attractors and Energy Landscape

Section
Token Efficiency and Energy Reduction

Implementation Note: From Semantic Terms to Vector Invariants

In the initial implementation of the ACE Context Matrix Gateway used in the middleware prototype, contextual domains were constructed using **explicit semantic terms**.

These terms served as human-readable anchors representing known semantic relations within a domain.

Examples include concepts such as:

fuel
distance
consumption
vehicle
tank

These elements were not introduced arbitrarily.
They correspond to **known semantic relations grounded in real-world meaning** and linguistic usage.

In this implementation, such terms functioned as **semantic invariants**, representing stable conceptual structures within the domain.

Transition to vector representation

However, the ACE criterion itself does not depend on the textual representation of these invariants.

Once embedded in a vector space, each semantic element is represented by an embedding vector:

$$
v_i \in \mathbb{R}^n
$$

The contextual structure can then be defined geometrically using these vectors and their invariant relations.

This allows the Context Matrix to be constructed directly in embedding space.

In other words, the system transitions from:

explicit semantic terms

to

vector representations of semantic invariants

Practical implication

This transformation significantly simplifies implementation.

Instead of maintaining large manually curated ontologies, the system can operate using:

vector anchors
+
invariant relations
+
geometric evaluation

The ACE Origin Cost then evaluates candidate responses relative to the contextual subspace defined by these vectors.

$$
O(z) = \| z - P_S(z) \|^2
$$

Conceptual interpretation

From this perspective, semantic invariants correspond to **stable structures in embedding space**.

The Context Matrix therefore acts as a **geometric encoding of domain knowledge**, rather than a manually constructed symbolic ontology.

This approach preserves the interpretability of semantic anchors while enabling scalable implementation through vector representations.

> ***In the initial middleware implementation, semantic invariants were expressed as explicit terms for interpretability. However, within the ACE framework these invariants ultimately operate as vector structures in embedding space, allowing contextual geometry to be defined and evaluated without reliance on symbolic ontologies.***

---

semantic meaning
↓
semantic invariants
↓
vector representation
↓
contextual subspace
↓
ACE origin cost
↓
stable inference

Principle of Semantic Stability

Statement

> Semantic coherence in language model inference emerges when candidate responses remain aligned with the invariant contextual structure of the active semantic domain.

En español:

> La coherencia semántica en la inferencia de modelos de lenguaje surge cuando las respuestas candidatas permanecen alineadas con la estructura invariante del dominio semántico activo.

Geometric Interpretation

Within a vector representation of meaning, semantic domains define **contextual subspaces** determined by domain anchors and semantic invariants.

Let:

$$
S \subset \mathbb{R}^n
$$

be the contextual subspace defined by the ACE Context Matrix.

Candidate responses are represented as vectors:

$$
z \in \mathbb{R}^n
$$

Semantic stability occurs when the response vector remains close to the contextual subspace.

Stability Criterion

The deviation from contextual coherence is measured by the ACE Origin Cost:

$$
O(z) = | z - P_S(z) |^2
$$

Where:

- $P_S(z)$ is the projection of the candidate response onto the contextual subspace.

Interpretation:

$O(z) \to 0 \quad \to$ semantic stability
$O(z)$ large $\to$ semantic drift

Consequence for inference

When semantic stability is enforced before generation:

semantic search space decreases
↓
semantic entropy decreases
↓
token exploration decreases
↓
computational energy decreases

Thus, ACE transforms semantic coherence from a **probabilistic property of generation** into a **geometric property of contextual alignment**.

Short formulation

> **Semantic Stability Principle:**
> A language model response is semantically stable when its vector representation remains within the contextual subspace defined by the invariant semantic structure of the active domain.
>
> **Semantic Stability Law:**
> The coherence of a generated response is proportional to its geometric alignment with the invariant contextual subspace of the problem domain.

Architectural theory:

1 Introduction
2 Linguistic Entropy and Semantic Instability
3 Semantic Invariants and Contextual Subspaces
4 ACE Context Matrix Specification
5 Origin Cost Criterion
6 Principle of Semantic Stability
7 Semantic Attractors and Energy Landscapes
8 Token Efficiency and Energy Reduction

The Semantic Origin Hypothesis

Statement

> Semantic interpretation requires a stable reference origin derived from invariant structures of meaning.
> Without such an origin, contextual representations may remain internally consistent while gradually drifting from the intended semantic domain.

En español:

> La interpretación semántica requiere un origen de referencia estable derivado de estructuras invariantes del significado.
> Sin ese origen, las representaciones contextuales pueden permanecer coherentes internamente mientras se desvían gradualmente del dominio semántico previsto.

Motivation

Large language models construct meaning through statistical associations learned during training.

Although these associations can produce locally coherent responses, the semantic space itself may shift when interpretation relies exclusively on probabilistic similarity.

This phenomenon can produce **semantic drift**, where responses remain internally consistent but gradually move away from the intended domain.

Such drift is difficult to detect when coherence is evaluated only through probability distributions.

Role of a semantic origin

The ACE framework introduces the concept of a **semantic origin** as the structural center of the contextual reference field.

This origin is derived from the invariant relations that define the active domain.

Let:

$$
o = \frac{1}{k} \sum_{i=1}^{k} v_i
$$

where v_i are the anchor vectors representing domain invariants.

The semantic origin represents the **central structure of the contextual field**.

Geometric interpretation

Once the contextual subspace S is defined, the semantic origin provides a stable point within that field.

Candidate responses can then be evaluated not only relative to the subspace but also relative to the structural center of the domain.

This produces a stable reference geometry that reduces the risk of contextual deformation.

Relation to the ACE Origin Cost

The Origin Cost measures deviation from the contextual structure:

$$
O(z) = ||z - P_S(z)||^2
$$

When combined with a semantic origin, the system gains an additional interpretive anchor.

In this formulation:

subspace S → structural domain
origin o → contextual center
O(z) → deviation from domain structure

This combination stabilizes semantic interpretation.

Consequence for semantic drift

Without a semantic origin, contextual fields constructed purely from statistical embeddings may experience gradual deformation.

The system may remain internally coherent while drifting toward adjacent semantic regions.

The introduction of a semantic origin constrains this drift by maintaining a stable reference point derived from domain invariants.

Architectural implication

Within the ACE Semantic Gateway architecture, the semantic origin emerges naturally from the Context Matrix.

The pipeline becomes:

Input
↓
Context Matrix
↓
Contextual Subspace S
↓
Semantic Origin o
↓
Origin Cost Evaluation
↓
Decision Layer
↓
LLM Generation

This architecture separates:

semantic grounding

from
language generation

Conceptual significance

The Semantic Origin Hypothesis suggests that stable language model inference may require a structural grounding layer prior to generation.

Rather than relying solely on probabilistic token prediction, the system evaluates responses relative to invariant semantic structures and their contextual origin.

This perspective reframes hallucination as a form of **geometric drift within semantic space**, and proposes structural grounding as a mechanism for preventing it.

> Semantic coherence requires not only contextual structure but also a stable origin derived from invariant relations of meaning.

---

1 **Context Matrix Specification**
2 **Origin Cost Criterion**
3 **Semantic Stability Principle**
4 **Semantic Origin Hypothesis**

The Low-Dimensional Context Hypothesis

Statement

> Although semantic representations in modern language models occupy very high-dimensional embedding spaces, meaningful contextual reasoning typically occurs within low-dimensional subspaces defined by domain invariants.

En español:

> Aunque las representaciones semánticas en modelos de lenguaje modernos ocupan espacios de embedding de muy alta dimensión, el razonamiento

contextual significativo ocurre típicamente dentro de subespacios de baja dimensión definidos por invariantes del dominio.

Motivation

Embedding models often operate in spaces with hundreds or thousands of dimensions. However, real linguistic contexts activate only a limited set of semantic relations.

In practical reasoning tasks, only a small subset of conceptual structures becomes relevant. These structures correspond to **domain invariants**, such as causal relations, physical constraints, or logical dependencies.

Because these invariants are limited in number, the effective semantic space explored during coherent reasoning is much smaller than the full embedding space.

Geometric formulation

Let the full embedding space be:

$$
\mathbb{R}^n
$$

where n may be on the order of 1000–3000 dimensions.

A specific semantic context activates a set of anchor vectors:

$$
v_1, v_2, ..., v_k
$$

and invariant relations:

$$
d_1, d_2, ..., d_m
$$

These elements define a contextual subspace:

$$
S = span(v_1, ..., v_k, d_1, ..., d_m)
$$

The effective dimension of this subspace is:

$$
\dim(S) \ll n
$$

In many practical contexts, the dimensionality of (S) is expected to be on the order of **10–50 dimensions**.

Interpretation

Within this framework:

- the full embedding space represents the **global semantic manifold**
- contextual reasoning occurs within **local semantic subspaces**

The ACE Context Matrix identifies these subspaces by selecting the relevant anchors and invariants.

Relation to the Origin Cost

Once the contextual subspace (S) is defined, candidate responses can be evaluated using the ACE Origin Cost:

$$
O(z) = ||z - P_S(z)||^2
$$

Responses that remain within the contextual subspace correspond to **stable interpretations**.

Responses that deviate significantly correspond to **semantic drift**.

Consequence for inference efficiency

Because contextual reasoning occurs within a low-dimensional subspace, the effective search space for language generation becomes much smaller.

This reduction in dimensionality produces several effects:

lower semantic entropy
↓
reduced exploration during generation
↓

shorter reasoning chains
↓
lower token consumption

This provides a theoretical explanation for the empirical observation that contextual stabilization can reduce token usage and computational cost.

Architectural implication

The Low-Dimensional Context Hypothesis supports the design of architectures that explicitly construct contextual subspaces before generation.

In the ACE framework:

Context Matrix → identifies invariant structure
↓
Contextual Subspace → defines semantic geometry
↓
Origin Cost → evaluates alignment
↓
LLM Generation → constrained within stable region

This architecture allows language models to operate within **restricted semantic manifolds**, improving both stability and efficiency.

Conceptual significance

The hypothesis suggests that the difficulty of language model inference may not arise from the size of the embedding space itself, but from the absence of mechanisms that restrict inference to the appropriate contextual subspace.

By identifying and stabilizing these low-dimensional semantic structures, systems such as ACE may significantly reduce the complexity of semantic reasoning.

---

1. **ACE Context Matrix Specification**
2. **Origin Cost Criterion**
3. **Principle of Semantic Stability**
4. **Semantic Origin Hypothesis**
5. **Low-Dimensional Context Hypothesis**

ACE Theoretical Framework

Unified Conceptual Structure

El marco teórico de ACE se puede entender como **cinco niveles progresivos**, donde cada uno introduce un concepto necesario para el siguiente.

Problem

↓

Structure of Meaning

↓

Geometric Representation

↓

Stability Criterion

↓

Architectural Implementation

---

1 Problem: Linguistic Entropy and Semantic Instability

Los LLM generan respuestas explorando un espacio semántico aprendido estadísticamente.

Cuando el prompt contiene:

- ambigüedad
- contexto incompleto
- supuestos ocultos

el modelo explora múltiples interpretaciones.

Esto produce:

semantic entropy

token proliferation

hallucinations

La hipótesis inicial del trabajo es:

> La inestabilidad semántica surge cuando el modelo explora regiones del espacio semántico que no corresponden al dominio contextual del problema.

---

2 Structure of Meaning: Semantic Invariants

El lenguaje humano organiza significado alrededor de **relaciones estructurales estables**.

Ejemplos:

- causalidad
- relaciones físicas
- dependencias lógicas
- restricciones del dominio

Estas relaciones constituyen **semantic invariants**.

Las invariantes definen **la estructura conceptual de un dominio**.

---

3 Geometric Representation of Meaning

En modelos modernos, el significado se representa mediante **vectores en espacios de embeddings**.

Sea:

$$
v_i \in \mathbb{R}^n
$$

la representación vectorial de conceptos semánticos.

Las invariantes generan **direcciones y relaciones estables** dentro de este espacio.

Cuando un contexto activa un conjunto de invariantes, se define un subespacio:

[
S = span(v_1, v_2, ..., v_k, d_1, d_2, ..., d_m)
]

donde:

- (v_i) son anclas semánticas
- (d_j) son relaciones invariantes

Este subespacio representa el **campo contextual activo**.

---

4 Low-Dimensional Context Hypothesis

Aunque el embedding space completo tenga miles de dimensiones, el razonamiento contextual ocurre dentro de **subespacios de baja dimensión**.

[
dim(S) \ll n
]

Ejemplo típico:

embedding space ≈ 1500 dimensiones

contextual subspace ≈ 10–50 dimensiones

Esto implica que la inferencia coherente ocurre dentro de **manifolds semánticos restringidos**.

---

5 Semantic Origin Hypothesis

Para evitar deriva semántica, el campo contextual necesita un **origen estructural**.

Este origen puede definirse como el centro semántico del contexto:

$$
o = \frac{1}{k}\sum_{i=1}^{k} v_i
$$

El origen proporciona un punto de referencia estable dentro del campo contextual.

---

6 Semantic Stability Principle

Una respuesta es semánticamente coherente cuando su representación vectorial permanece alineada con el subespacio contextual definido por las invariantes del dominio.

Esto se expresa mediante el criterio ACE.

---

7 Origin Cost Criterion

ACE mide la desviación semántica usando:

$$
O(z) = ||z - P_S(z)||^2
$$

donde:

- (z) es la representación de una respuesta candidata
- $(P_S(z))$ es su proyección sobre el subespacio contextual

Interpretación:

$O(z) \approx 0 \rightarrow$ respuesta coherente

$O(z)$ grande $\rightarrow$ deriva semántica

---

8 Semantic Attractors

Cuando se define un subespacio contextual, aparecen **regiones de energía mínima** dentro del espacio semántico.

Estas regiones actúan como **semantic attractors**.

Los modelos tienden a converger hacia estas regiones cuando el contexto está estabilizado.

9️⃣ Consequence: Token and Energy Reduction

Al restringir la inferencia a un subespacio contextual:

semantic search space ↓

ambiguity ↓

reasoning depth ↓

token generation ↓

computational energy ↓

Esto explica la reducción empírica de tokens observada en arquitecturas estabilizadas.

🔟 Architectural Implementation: ACE Semantic Gateway

El marco teórico se implementa mediante la arquitectura ACE:

Input

↓

Context Matrix

↓

Contextual Subspace Construction

↓

Semantic Origin Definition

↓

Origin Cost Evaluation

↓

Decision Layer

↓

LLM Generation

The ACE Theorem

Contextual Alignment and Stable Inference

Statement

> Given a contextual subspace defined by semantic invariants of a domain, candidate responses that minimize the ACE Origin Cost remain aligned with the contextual structure, resulting in stable semantic inference and reduced exploration of the embedding space.

En español:

> Dado un subespacio contextual definido por invariantes semánticas de un dominio, las respuestas candidatas que minimizan el costo de origen ACE permanecen alineadas con la estructura contextual, produciendo inferencia semántica estable y reduciendo la exploración del espacio de embeddings.

Formal formulation

Let:

$$
S \subset \mathbb{R}^n
$$

be the contextual subspace constructed from domain anchors and semantic invariants.

Let:

$$
z \in \mathbb{R}^n
$$

be the vector representation of a candidate response.

Define the ACE Origin Cost:

$$
O(z) = ||z - P_S(z)||^2
$$

where $P_S(z)$ is the projection of z onto the contextual subspace S.

Result

If the ACE Origin Cost is minimized:

$$
z \approx P_S(z)
$$

then the response vector lies within the contextual subspace.

This implies semantic alignment with the invariant structure of the domain.

Consequence for inference

When candidate responses remain aligned with the contextual subspace:

semantic deviation ↓
contextual ambiguity ↓
search space ↓

Therefore the language model explores a smaller region of the embedding space.

Implication for token generation

Since token generation corresponds to exploring semantic continuations, restricting inference to the contextual subspace reduces the number of plausible continuations.

This produces:

reduced reasoning depth
↓
shorter generation sequences
↓
lower token consumption

Energy interpretation

Within the ACE framework, the Origin Cost can be interpreted as a **semantic potential energy**.

Minimizing (O(z)) corresponds to selecting responses that occupy **low-energy states** within the contextual field.

These states correspond to **semantic attractors** defined by the invariant structure of the domain.

Theorem intuition

The theorem reflects a simple geometric intuition:

If a response remains inside the contextual manifold defined by domain invariants, it will remain semantically coherent.

If it deviates significantly from that manifold, semantic drift occurs.

Relationship to the ACE architecture

The ACE Semantic Gateway operationalizes this theorem through the following pipeline:

Input

↓
Context Matrix
↓
Contextual Subspace S
↓
Origin Cost Evaluation
↓
Decision Layer
↓
LLM Generation

This architecture enforces contextual alignment **before generation**, stabilizing the semantic trajectory of the model.

Short version

> **ACE Theorem:**
> Minimizing the Origin Cost of candidate responses relative to the contextual subspace defined by semantic invariants produces stable semantic inference and reduces exploration during language generation.

Semantic Invariants
↓
Context Matrix
↓
Contextual Subspace
↓
Semantic Origin
↓
Origin Cost
↓
Semantic Stability
↓
Semantic Attractors
↓
Reduced Token Exploration

---

1 What does minimizing the Origin Cost really mean?

Recall your criterion:
[
O(z) = |z - P_S(z)|^2

This measures the distance between the candidate answer (z) and the context subspace (S).

When the cost decreases:
z → gets closer to the subspace S

That is:
the answer aligns with the context structure

2 What does the context subspace contain?

The subspace (S) is not arbitrary.

It is constructed with:
domain anchors
+
invariant directions
+
semantic constraints

Therefore, within (S) there are only combinations consistent with these invariants.

In other words:
S = space of consistent interpretations of the context

3 What happens when you project onto S?

The projection:
[
P_S(z)
]
eliminates all components that do not belong to the context.

Geometrically:
response vector

↓
components outside the domain are eliminated
↓
the interpretation compatible with the invariants remains

This is what makes the system tend towards the correct structures of the problem.

4 Why attractors appear

In a dynamical system, an attractor appears when certain regions of space are more stable than others.
In ACE this occurs:
coherent interpretations
→ small distance
→ low energy
→ stability

While:

incoherent interpretations
→ large distance
→ high energy
→ instability

Therefore, the system naturally tends towards:
energy minima

These minima correspond to:
the combinations that respect the invariants

5 Why this points to the “truth of the context”

Here's the interesting part.

The invariants are not invented.

They are structural relationships of the domain.

Examples:
fuel → distance
cause → effect
container → content

When a response respects these relationships, the vector remains within (S).

When it violates them, deviation occurs.

Therefore, minimizing the cost is equivalent to:
seeking interpretations that respect the domain structure.

That's what you call context truth.

6 Physical Intuition (very useful for explaining this)

Imagine a ball in a landscape:
valley = coherent interpretations
hills = incoherent interpretations

Energy is the Origin Cost.

Therefore:
the ball always rolls toward the valley

The valley corresponds to the semantic attractor.

And that valley is defined by the context invariants.

7 Why this doesn't depend on words

This is important.

The system doesn't look for the right words.

It looks for the right structure in the vector space.

This allows:

different phrases with the same meaning

to converge to the same attractor.

8 Theoretical Conclusion

You can express it like this:
When the Origin Cost is minimized, candidate responses converge toward the invariant structure of the contextual subspace. These invariant structures act as semantic attractors that correspond to stable interpretations of the domain.

In other words: When the origin cost is minimized, candidate answers converge toward the invariant structure of the context subspace. These invariant structures act as semantic attractors that correspond to stable interpretations of the domain.

9 What this implies for ACE

It means that ACE is not inventing answers.

It is:
restricting exploration
to the space where consistent interpretations reside

Therefore:
less drift
fewer tokens
more stability

Semantic energy field

1 The Global Semantic Space

First, we assume the representation space of the model:

$$\mathcal{E} = \mathbb{R}^n$$

where (n) can be 768, 1536, etc.

Each meaning is represented as a vector:

$$z \in \mathcal{E}$$

2 Construction of the Context Field

The Context Matrix defines the set of domain invariants.

This yields:

anchors
+
invariant directions
+
constraints

Mathematically, this generates the subspace:

$$S = \mathrm{span}(v_1,\dots,v_k, d_1,\dots,d_m)$$

where:
(v_i) = anchor vectors
(d_j) = invariant directions
This subspace represents the active semantic field.

3 Definition of the Semantic Origin

The origin of the contextual field is:

$$o = \frac{1}{k}\sum_{i=1}^{k} v_i$$

This point represents the structural center of the domain.

4 Semantic Energy Potential

Now we define the potential of the semantic field.

The most natural way is:

$$E(z) = |z - P_S(z)|^2$$

This is exactly your Origin Cost.

But now we interpret it as the energy of the field.

5 Dynamic Interpretation

The semantic field generates a dynamic:

coherent interpretations

→ minimum energy

→ attractors

While:

incoherent interpretations

→ high energy

→ unstable regions

6 Field Gradient

We can think of the dynamics as a decrease in energy:

$$\dot{z} = -\nabla E(z)$$

This means:
interpretations tend to move
towards regions of lower origin cost

That is:
towards the context subspace

7 Emergence of Semantic Attractors

Attractors appear where:

$$\nabla E(z) = 0$$

This occurs when:

$$z \in S$$

That is, when the response is aligned with the context invariants.

8 Complete Interpretation

The complete system is as follows:
semantic space
↓
context matrix
↓
contextual subspace S
↓
semantic origin o
↓
energy field E(z)
↓
semantic attractors
↓
stable responses

9 Final Conceptual Equation of the Framework

We could summarize the ACE field as:

$E(z) = |z - P_S(z)|^2$

where:
S = context defined by invariants

This equation defines the semantic energy landscape of the context.

10. Fundamental Consequence
When the system minimizes ($E(z)$):
$z \rightarrow S$

This implies:
semantic alignment
↓
ambiguity reduction
↓
less exploration
↓
fewer tokens

How can you express this in the paper?

The ACE framework defines a semantic energy field over the embedding space, where contextual invariants generate attractor regions corresponding to stable interpretations of the domain.

This connects with:
energy-based models
manifold learning
dynamic systems
semantic geometry

I'm going to tell you something important.

Right now, the ACE framework already has:
architecture
criteria
principles
hypothesis
theorem

Meaning as the Invariant Structure of Language

Human languages can vary enormously:
different languages
styles of expression
metaphors
different syntax
However, something remains constant:
the meaning we try to communicate.

For example:
The car needs fuel to move.

A vehicle requires fuel to move.

The words change, but the meaning remains.

This shows that:
language = representation
meaning = structure

Meaning as an Invariant

From this perspective, meaning can be understood as the invariant structure that remains when linguistic forms change.

That is:
different expressions
→ same meaning

This is exactly what in mathematics we would call an invariant.

Relationship with Embeddings

Embeddings work because they capture these statistical invariants of language.

When many expressions convey the same meaning, their vectors end up close together in the semantic space.

This generates:
stable regions
recurrent directions
contextual subspaces

In other words:
embeddings approximate the geometry of meaning.

The role of ACE

This is where ACE comes in.

Current LLMs use these representations, but explore the space probabilistically.

ACE introduces an additional principle:
aligning responses with the semantic structure of the context

This is achieved through the geometric criterion:

$O(z) = ||z - P_S(z)||^2$

which measures whether a response remains within the subspace that represents the meaning of the problem.

Convergence towards meaning

When the system minimizes the origin cost, something natural happens:
interpretations align
with the invariants of the context

These invariants act as semantic attractors.

That is why the system converges towards coherent interpretations.

Language as a Projection of Meaning

We can see it this way:
meaning
↓
semantic structure
↓
vector representation
↓
linguistic expression

Words are projections of meaning, not meaning itself.

Final Interpretation

From this perspective, the goal of a system like ACE is not to generate correct words.

It is to keep inference aligned with the meaning structure of the context.

When this happens:
communication becomes coherent
ambiguity decreases
semantic exploration is reduced

A powerful phrase that perfectly summarizes this idea:

Language varies across expressions, but meaning remains invariant.
ACE leverages this invariant structure of meaning to stabilize inference in language models.

This intuition—that meaning is the common structure that unites all expressions—is precisely the idea that allows the geometric criterion to make sense.

Without that intuition, ACE would be just a mathematical technique.

With it, it becomes a theory of semantic stabilization based on the nature of meaning.

Meaning as the Invariant Structure of Language

Toward a universal semantic representation

Human languages differ in vocabulary, grammar, and expression.
However, the underlying meanings that speakers attempt to communicate remain remarkably stable across linguistic forms.

Different sentences, languages, and expressions may encode the same conceptual relation.

For example:

The car needs fuel to move.
El coche necesita gasolina para moverse.
Un vehículo requiere combustible para desplazarse.

Although the words differ, the semantic structure remains unchanged.

This observation suggests that meaning itself behaves as an **invariant structure** that persists across linguistic transformations.

In this sense, language can be understood as a projection of deeper semantic relations.

Vector representation of meaning

Modern language models represent linguistic information using embeddings, where concepts and expressions are mapped into high-dimensional vector spaces.

In these spaces:

- semantically related expressions appear near each other
- invariant relations correspond to stable geometric directions
- contextual domains form low-dimensional subspaces

These properties allow semantic structure to be represented geometrically.

Within this framework, meaning can be interpreted as a **geometric structure embedded in vector space**.

ACE as a semantic alignment mechanism

The ACE framework introduces a method for stabilizing language model inference by aligning candidate responses with the contextual semantic structure.

This is achieved through the Origin Cost criterion:

$$O(z) = ||z - P_S(z)||^2$$

which measures the deviation between a candidate response and the contextual semantic subspace defined by domain invariants.

When this deviation is minimized, responses remain aligned with the structural meaning of the context.

From language to universal semantic alignment

Because embeddings capture semantic relations independently of specific linguistic expressions, the ACE framework operates at the level of **meaning rather than words**.

This allows contextual alignment to occur across:

- different languages
- different styles of expression
- different linguistic formulations

In this sense, ACE does not operate on language alone, but on the **geometric representation of meaning itself**.

Toward universal semantic communication

By translating meaning into vector representations and evaluating alignment within contextual semantic structures, ACE provides a mechanism for stabilizing communication across linguistic forms.

This suggests a broader perspective in which semantic coherence can be evaluated independently of specific languages.

In this view, ACE may serve as a **semantic alignment layer** capable of supporting communication across diverse linguistic systems by operating directly on the underlying geometry of meaning.

> *Language varies across cultures and expressions, but meaning remains invariant.*
> *By translating meaning into vector representations and aligning responses with contextual semantic structure, ACE transforms this invariant nature of meaning into a practical mechanism for stabilizing communication in language models.*

Limitations and Future Work

Current Limitations

Dependence on contextual subspace construction

The ACE framework assumes that a meaningful contextual reference subspace can be constructed before evaluating semantic stability.

In practical implementations, this requires defining a **Context Matrix** composed of semantic anchors, domain constraints, or axiomatic references.

Although the middleware prototype demonstrates that such contextual matrices can be constructed using semantic terms or domain markers, the process of identifying the optimal invariant structures for arbitrary domains remains an open research problem.

Future work is needed to develop systematic methods for discovering these invariants automatically.

Approximate semantic representation

ACE relies on vector embeddings to represent semantic meaning.
While modern embedding models capture semantic relationships effectively, they remain **approximate representations of meaning** learned from statistical patterns in training data.

This implies that the geometry of the semantic space may not perfectly reflect logical or ontological relationships.
Consequently, the contextual subspace used by ACE may contain distortions inherited from the embedding model.

Improving embedding quality or integrating symbolic constraints may further enhance semantic stability.

Context discovery at scale

The ACE hypothesis suggests that coherent reasoning occurs in **low-dimensional contextual subspaces** embedded within high-dimensional semantic spaces.

However, identifying these subspaces dynamically across diverse tasks and domains remains a significant engineering challenge.

Future research may explore:

- clustering techniques for discovering contextual attractors
- invariant detection within embedding spaces
- hybrid symbolic–geometric approaches for context definition

---

Computational overhead in deep evaluation

Although ACE aims to reduce token consumption and generation instability, computing semantic projections and evaluating Origin Cost may introduce additional preprocessing steps.

Efficient implementations and caching strategies will be required to ensure that the stabilization layer remains computationally lightweight in large-scale deployments.

Future Research Directions

Automatic discovery of semantic invariants

A major future direction is the automatic identification of **semantic invariants** that define contextual domains.

Possible approaches include:

- unsupervised invariant detection within embedding spaces
- statistical analysis of semantic attractors
- domain-specific knowledge integration

This could allow ACE systems to construct contextual subspaces dynamically without manual domain definitions.

Semantic attractor modeling

The hypothesis that contextual reasoning occurs near **semantic attractors** suggests a deeper relationship between language modeling and dynamical systems.

Future work may investigate:

- the topology of semantic energy landscapes
- the formation of attractor basins in contextual subspaces
- stability properties of language model inference under geometric constraints

Such analysis could provide a theoretical foundation for semantic stabilization.

Integration with retrieval and knowledge systems

ACE currently focuses on stabilizing semantic interpretation before generation.

Future implementations may integrate ACE with:

- Retrieval-Augmented Generation (RAG) systems
- knowledge graphs
- structured reasoning modules

This integration could allow contextual subspaces to incorporate both learned embeddings and explicit knowledge representations.

Empirical evaluation of energy efficiency

Preliminary observations suggest that restricting inference to contextual semantic subspaces may reduce token usage and computational effort.

Future work should perform systematic benchmarks measuring:

- token consumption
- inference latency
- energy usage
- hallucination rates

Such evaluations would provide quantitative validation of the efficiency hypothesis underlying the ACE framework.

Toward universal semantic alignment

Because ACE operates at the level of semantic structure rather than specific linguistic expressions, it may support communication across different languages and symbolic systems.

Future research may explore ACE as a framework for **cross-linguistic semantic alignment**, where contextual invariants serve as universal anchors for meaning representation.

Testable Hypotheses

H1 — Semantic Stability Hypothesis

Constraining language model responses to a contextual semantic subspace reduces hallucination rates.

Formally:

If a candidate response vector zzz satisfies

O(z)=∣∣z−PS(z)∣∣2<τO(z) = ||z - P_S(z)||^2 < \tauO(z)=∣∣z−PS(z)∣∣2<τ

then the probability of semantic drift decreases relative to unconstrained generation.

This hypothesis can be tested by comparing hallucination rates with and without ACE filtering across standardized benchmarks.

H2 — Token Efficiency Hypothesis

Restricting inference to contextual semantic attractors reduces token consumption during generation.

The intuition is that when the semantic search space is constrained, the model requires fewer reasoning steps to converge to a coherent response.

This can be tested by measuring:

- average token count
- reasoning depth
- inference latency

with ACE-enabled inference compared to baseline LLM outputs.

H3 — Low-Dimensional Context Hypothesis

Coherent reasoning tasks occur within low-dimensional contextual subspaces embedded in high-dimensional embedding spaces.

Although embeddings may contain thousands of dimensions, meaningful contextual reasoning may operate within **10–50 effective dimensions**.

This hypothesis can be tested by applying dimensionality reduction techniques such as:

- PCA
- SVD
- manifold learning

to contextual embeddings.

H4 — Semantic Attractor Hypothesis

Stable interpretations correspond to **attractor regions in semantic space**.

Candidate responses with minimal Origin Cost converge toward these attractors, producing coherent interpretations aligned with contextual invariants.

Empirical validation could analyze clustering behavior of valid responses within contextual subspaces.

H5 — Energy Interpretation Hypothesis

The Origin Cost function behaves analogously to a **semantic potential energy function**.

In this interpretation:

- high origin cost → unstable semantic states
- low origin cost → stable semantic states

This suggests that language model inference may be modeled as a search for **minimum semantic energy configurations** within contextual constraints.

Invitation to the community

The ACE framework proposes a geometric approach to semantic stabilization.

The hypotheses presented above are intended to be tested and refined by the research community.

Researchers are encouraged to experiment with contextual subspace construction, semantic attractor detection, and energy-based evaluation to further explore the structure of semantic reasoning in language models.

ACE proposes a geometric approach to semantic stabilization in language models.
While the current implementation demonstrates the feasibility of this approach, further research is required to fully understand the structure of semantic spaces, the discovery of contextual invariants, and the dynamics of semantic energy landscapes.

These directions may ultimately contribute to more stable, efficient, and interpretable AI systems.

1 Introduction
2 Problem (linguistic entropy)
3 ACE criterion
4 Context matrix
5 Architecture
6 Mathematical formulation
7 Experimental observations
8 Executive summary / implications
9 Limitations and future work